

Solvency Field Theory V13.9

Locally Real Without Bell Locality:
How Shared Written History
Produces Quantum Correlations

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SFT does not escape Bell by making particles unreal, settings predetermined, probabilities negative, or signals faster than light. It keeps local interactions and causal records, but treats the prepared entangled source as one irreducible relational support object whose actual-context joint sectors cannot be decomposed into a measurement-independent mixture of separate local answer tables.

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1 The narrow question

This paper does not claim that SFT has derived all of quantum mechanics. It asks one specific question:

Can locally real supports and irreversible local records reproduce the singlet correlation law without inserting $-\cos\theta$, prewriting all counterfactual answers, signaling, preferred measurement order, measurement dependence, or postselection?

Bell’s theorem says no — if “locally real” means Bell-local factorization:

$$P(A, B \mid a, b) = \int P(A \mid a, \lambda) P(B \mid b, \lambda) d\rho(\lambda). \quad (1)$$

Any such model satisfies $|S_{\text{CHSH}}| \leq 2$. Experiments find $|S| = 2\sqrt{2}$.

SFT’s answer: the particles are locally real, the measurements are local, the records are irreversible, and no signal passes between the wings. But the prepared source is one relational object that cannot be decomposed into a mixture of separate local answer tables. The particles are real. The relation between them is real. The relation is not separable. That is enough to violate Bell’s inequality without violating causality.

2 What Bell actually assumes

Bell’s theorem requires four ingredients working together:

Realism: outcomes exist as definite values.

Locality: outcome A depends only on setting a and source variable λ , not on the distant setting b .

Measurement independence: the source variable λ does not depend on which settings will be chosen: $P(a, b \mid \lambda) = P(a, b)$.

Separability: the joint probability factorizes into a product of two single-wing functions.

SFT preserves the first three. It rejects the fourth. The source object Π_0 is nonseparable — not because information travels between the particles, but because their shared formation history created a relational constraint that no product of local descriptions can reproduce.

3 The source relation

Two photons collide with enough invariant mass to produce a particle–antiparticle pair. Charge conservation balances the holonomy windings: $\theta_g^- + \theta_g^+ = 0$. Opposite charge is automatic. A specific spin state is not.

The singlet — total spin zero — is a *prepared channel*, selected by the angular-momentum structure of the formation event. It is not automatic for every pair. The derivation canon conditions on this prepared channel. The microscopic source-production operator that determines the formation branching fractions remains an explicit open debt.

For the selected pure $L=0$, $S=0$ channel, the source support projects onto a one-dimensional sector of the two-particle space:

$$\Pi_0 = \frac{1}{4}(I - \sigma_A \cdot \sigma_B). \quad (2)$$

This object has Schmidt rank 2: it cannot be written as a product of single-particle states. The partial-transpose test confirms genuine entanglement. And the two single-particle restrictions are both $I/2$ — neither particle leaves the source with a readable preferred orientation.

The source is definite as a pair. It is maximally uncommitted as individuals.

4 Local questions, joint support

Each detector asks one binary question along a chosen axis. The particle arriving at detector A has a 4π spinor closure with a two-dimensional local support factor. On a two-dimensional space, the most general Hermitian signed analyzer is $D(n) = d_0 I + \mathbf{d} \cdot \boldsymbol{\sigma}$. Rotation covariance forces the response vector along the axis: $\mathbf{d} = \kappa n$. Axis reversal removes the identity bias. Ideal sharpness fixes $|\kappa| = 1$. Therefore the local analyzer is uniquely

$$\boxed{D(n) = n \cdot \boldsymbol{\sigma}}, \quad (3)$$

and the two outcome projectors are

$$P_s(n) = \frac{1}{2}(I + s n \cdot \boldsymbol{\sigma}), \quad s = \pm 1. \quad (4)$$

On a two-state space, every polynomial in $X_n = n \cdot \boldsymbol{\sigma}$ collapses to $\alpha I + \beta X_n$ because $X_n^2 = I$. Higher-order detector terms do not exist. The linearity is exact, not perturbative.

The four joint-support sectors are:

$$Q_{st}(a, b) = P_s^A(a) \otimes P_t^B(b). \quad (5)$$

These commute because they act on separate factors: $P_s^A P_t^B = P_t^B P_s^A$. There is no measurement order.

The candidate-support weights are:

$$\mu_{st}(a, b) = \text{Tr}(\Pi_0 Q_{st}(a, b)). \quad (6)$$

Direct matrix multiplication of the independently derived source projector and detector projectors gives:

$$\boxed{\mu_{st}(a, b) = \frac{1}{4}[1 - st(a \cdot b)]}. \quad (7)$$

The angular dependence $a \cdot b$ was not inserted. It emerged from the composition of three objects derived in separate gates from separate physics. The source object knows nothing about detector axes. The detector projectors know nothing about the source state. The cosine appeared when they met.

5 From support to recorded frequency

The support weights μ_{st} measure geometric overlap between the source domain and each detector sector. To become detection probabilities, they must pass through an intensity bridge.

Let the eligible-write intensity in channel (s, t) be $I_{st} = F(\mu_{st})$. Disjoint legal micro-support must add: $F(x + y) = F(x) + F(y)$. With positivity, the unique solution is $F(x) = \kappa x$. An ideal calibrated detector has no intrinsic preference among its channels: swapping the signed labels of either detector maps the four sectors transitively, forcing a single common coefficient $\kappa_{st} = \kappa_{ab}$.

The write-race completion gives:

$$P(s, t \mid a, b) = \frac{I_{st}}{\sum_{u,v} I_{uv}} = \frac{\mu_{st}}{\sum_{u,v} \mu_{uv}} = \mu_{st}, \quad (8)$$

since $\sum_{s,t} \mu_{st} = 1$.

The ideal completed-event kernel is therefore:

$$\boxed{P(s, t \mid a, b) = \frac{1}{4} [1 - st (a \cdot b)]}. \quad (9)$$

The correlation function:

$$E(a, b) = \sum_{s,t} st P(s, t \mid a, b) = -a \cdot b. \quad (10)$$

No quantum equation was imported. The probability law was derived from: the source formation mechanics (M1–M4), the detector closure structure (M5), the joint composition (M6), the contextual activation (M7), the additive intensity bridge (M8), and the frame-independent completion (M9).

6 Causal locality without Bell locality

The local record maps act on disjoint spacelike factors. They commute: $P_s^A P_t^B = P_t^B P_s^A$. Different Lorentz frames may reverse the coordinate-time order of the two detections, but the common-future ledger join is the same.

The local marginals:

$$\sum_t P(s, t \mid a, b) = \frac{1}{2}, \quad \sum_s P(s, t \mid a, b) = \frac{1}{2}. \quad (11)$$

Neither depends on the distant setting. No controllable signal passes between the wings. This is a structural theorem of the frozen kernel, not a numerical accident.

SFT preserves causal locality but is not Bell-local.

The source relation Π_0 is nonseparable. The joint support weights μ_{st} cannot be written as a measurement-independent mixture of products of local response functions. The polytope of all 64 deterministic local strategies was searched exhaustively: no nonnegative mixture reproduces the frozen kernel. The nearest local model misses by at least $1/32$ per probability cell.

7 The violation

For the canonical CHSH witness:

$$S = E(a_0, b_0) + E(a_0, b_1) + E(a_1, b_0) - E(a_1, b_1), \quad (12)$$

with predeclared angles $a_0 = 0^\circ$, $a_1 = 90^\circ$, $b_0 = 45^\circ$, $b_1 = -45^\circ$:

$$\boxed{|S| = 2\sqrt{2} \approx 2.8284}. \quad (13)$$

The Bell-local bound is $|S| \leq 2$. The violation margin is $2\sqrt{2} - 2 \approx 0.8284$. The witness angles were hashed before scoring. They saturate the geometric maximum of the frozen correlation — this is the Tsirelson bound, not an optimized post-hoc selection.

8 The ten attacks

The frozen model was subjected to ten hostile gates. No mechanism was changed after freezing.

H1: Factorization kill. All 64 deterministic local strategies enumerated. No nonnegative mixture reproduces the kernel. Infeasibility certified by linear programming.

H2: CHSH test. $|S| = 2\sqrt{2}$ exactly. Predeclared witness. Hashed before scoring.

H3: No-signaling stress. Raw local marginals are remote-setting-independent under arbitrary local detector noise, nulls, and latency. Postselection on coincidence timing can create apparent signaling from a perfectly clean kernel.

H4: Superdeterminism escape. A Bell-local model reproducing the kernel exists if measurement independence is relaxed by 0.046 bits of source–setting mutual information. The frozen SFT model does not use this escape: source and settings are structurally separated.

H5: Frame ordering. All six causal orderings of two spacelike detections produce identical joint predictions. No preferred first measurement. Selecting events by coordinate-time order manufactures apparent signaling from pure postselection.

H6: Delayed choice. Settings chosen after source formation, even after the remote detection in some frames, produce unchanged predictions. The source is a constraint, not a table of answers for settings that may not yet exist.

H7: Entanglement swapping. Two particles with no common formation history become entangled — conditioned on a middle measurement result — through a later local joint interaction that joins their independent ancestries. This forced a genuine ontology upgrade: entanglement is a heralded relational ledger sector, not only a birth-vertex property.

H8: GHZ. The three-party GHZ state produces Mermin $M = 4$ versus the local bound $|M| \leq 2$. All 729 three-party ternary strategies fail. The contradiction is deterministic, not statistical. This forced a second ontology upgrade: some relational objects are genuinely N -party and cannot be reconstructed from pairwise reductions.

H9: Decoherence. Environmental history overlap γ and source visibility V degrade the Bell witness smoothly. Entanglement can survive past the point where the specific CHSH witness stops violating. Exact thresholds derived for Bell and GHZ witnesses.

H10: Detection loophole. The exact detector efficiency threshold for loophole-free violation with the maximally entangled source is $\eta_{\text{crit}} = 2/(1 + \sqrt{2}) \approx 0.828$. Coincidence-normalized CHSH alone does not close the loophole. A complete experiment-entry contract is specified: trial definition, null records, raw singles, and pre-registered analysis.

9 What changed in SFT

The hostile testing produced two genuine discoveries, not cosmetic extensions.

The origin upgrade. Before H7, entanglement in SFT meant shared formation history from a common vertex. After H7, entanglement is a relational ledger sector that can be created by *any* local joint interaction joining previously independent histories. Birth is one route. A later joint measurement on partner particles is another. Neither requires a signal, a preferred frame, or retroactive rewriting of completed records.

The arity upgrade. Before H8, all SFT relational objects were pairwise. After H8, some prepared source channels require genuine N -party relational objects that cannot be reconstructed from their pairwise reductions. The GHZ state has zero pairwise entanglement across every pair, yet is genuinely tripartite-entangled. The relational support must live in the full N -party ledger.

Neither upgrade adds a particle, field, dimension, or superluminal mechanism.

10 Claim boundary

Claim	Status	Boundary
Singlet kernel	Derived	Conditional on prepared pure $S=0$ channel, ideal detectors, and measurement independence.
$S = 2\sqrt{2}$	Derived	Predeclared witness on frozen kernel.
No-signaling	Proved	Structural theorem of the kernel.
Bell-local exclusion	Proved	Under measurement independence.
Entanglement swapping	Architecture pass	Algebraic structure correct; microscopic mode matching open.
GHZ / N -party	Architecture pass	Mermin $M = 4$; source production open.
Detection threshold	Derived	$\eta > 2/(1+\sqrt{2})$ for maximally entangled source.
Source production	Open	Microscopic formation dynamics.
Detector amplification	Open	Macroscopic record objectivity.
Decoherence rates	Open	Environmental coupling dynamics.

The sentence

SFT preserves causal locality but is not Bell-local. The prepared source is one relational object. The two detectors ask local questions. The answers they find are constrained by the relation they share. The relation was written at formation and preserved by no-overwrite. It is not a signal. It is not a table of pre-written answers. It is a permanent geometric fact about two particles that were born together, carried forward by the same irreversibility that makes writes permanent, clocks tick, and things fall.

Acknowledgments

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References

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